Sprint 2 Submission
Project Name: ClosetSwap
Software Engineering (CMP-N204-0)
University of Roehampton
Group Name: The Group

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Introduction

Sprint 2 focuses on formalising the functional and structural requirements of the ClosetSwap platform prior to technical implementation. The purpose of this sprint is to ensure that system behaviour, data structures, and feature scope are clearly defined before development begins in Sprint 3.

ClosetSwap is a sustainable fashion marketplace designed to support community exchange of clothing items. The platform enables users to register, create listings, browse available items, and interact within a structured digital environment. This document presents the prioritised requirements, system modelling artefacts, database design, user interface planning, and Sprint 3 implementation specification.

The artefacts included demonstrate traceability between requirements and technical design, ensuring that development in the next sprint is grounded in clearly defined system logic and realistic scope control.

Requirements Definition and Prioritisation

The system requirements were defined using a user-centred approach informed by stakeholder roles including visitors, buyers, sellers, and administrators. Features were prioritised using the MoSCoW method in order to ensure incremental and realistic development.

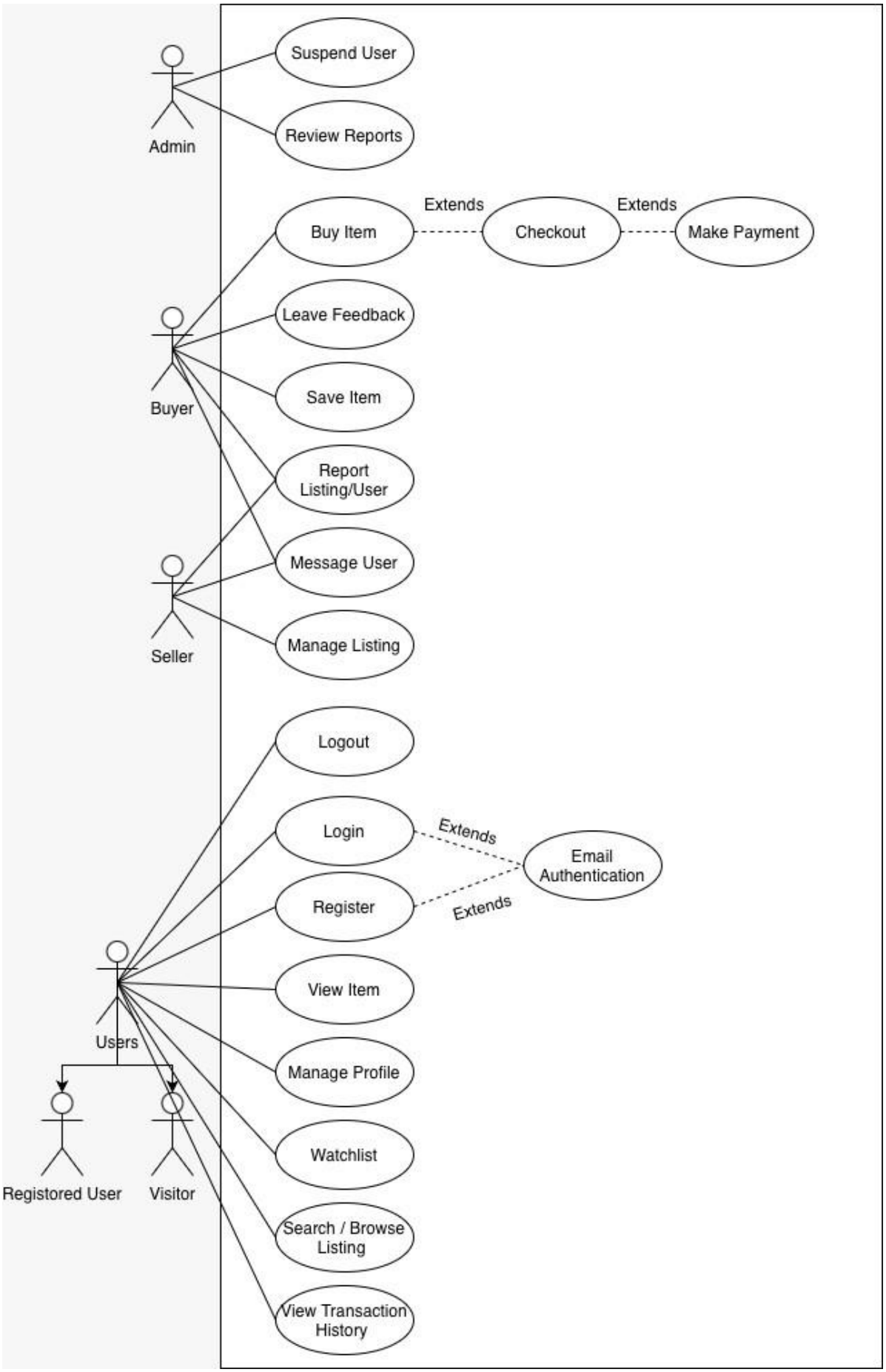
The following user stories define the functional scope of the application:

User Stories

Use Case Model

The use case model identifies the external actors interacting with the system and defines the functional capabilities offered within the platform boundary.

The primary actors include visitors, registered users, buyers, sellers, and administrators. The diagram captures interactions such as account registration, authentication, listing management, purchasing, messaging, and moderation actions. The model clarifies role-based access and highlights relationships between system features. Extension relationships illustrate dependent functionality such as checkout and payment processing.

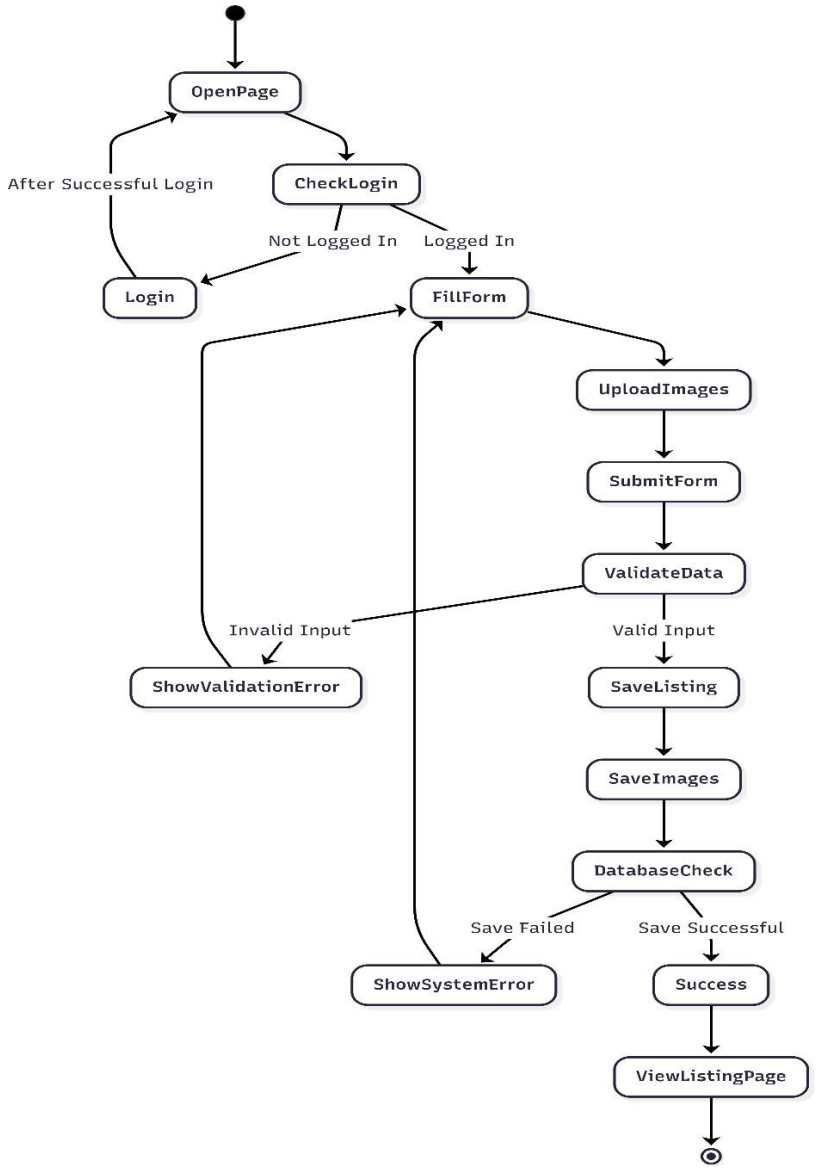

Admin
Suspend User
Review Reports
Buyer
Buy Item
Extends
Checkout
Extends
Make Payment
Leave Feedback
Save Item
Report Listing/User
Message User
Seller
Manage Listing
Logout
Login
Extends
Email Authentication
Register
Extends
View Item
Manage Profile
Watchlist
Search / Browse Listing
View Transaction History
Users
Registered User
Visitor

Behavioural Model – Activity Diagram

The activity diagram models the logical flow involved in creating a listing. It captures authentication verification, form submission, data validation, error handling, and successful database persistence.

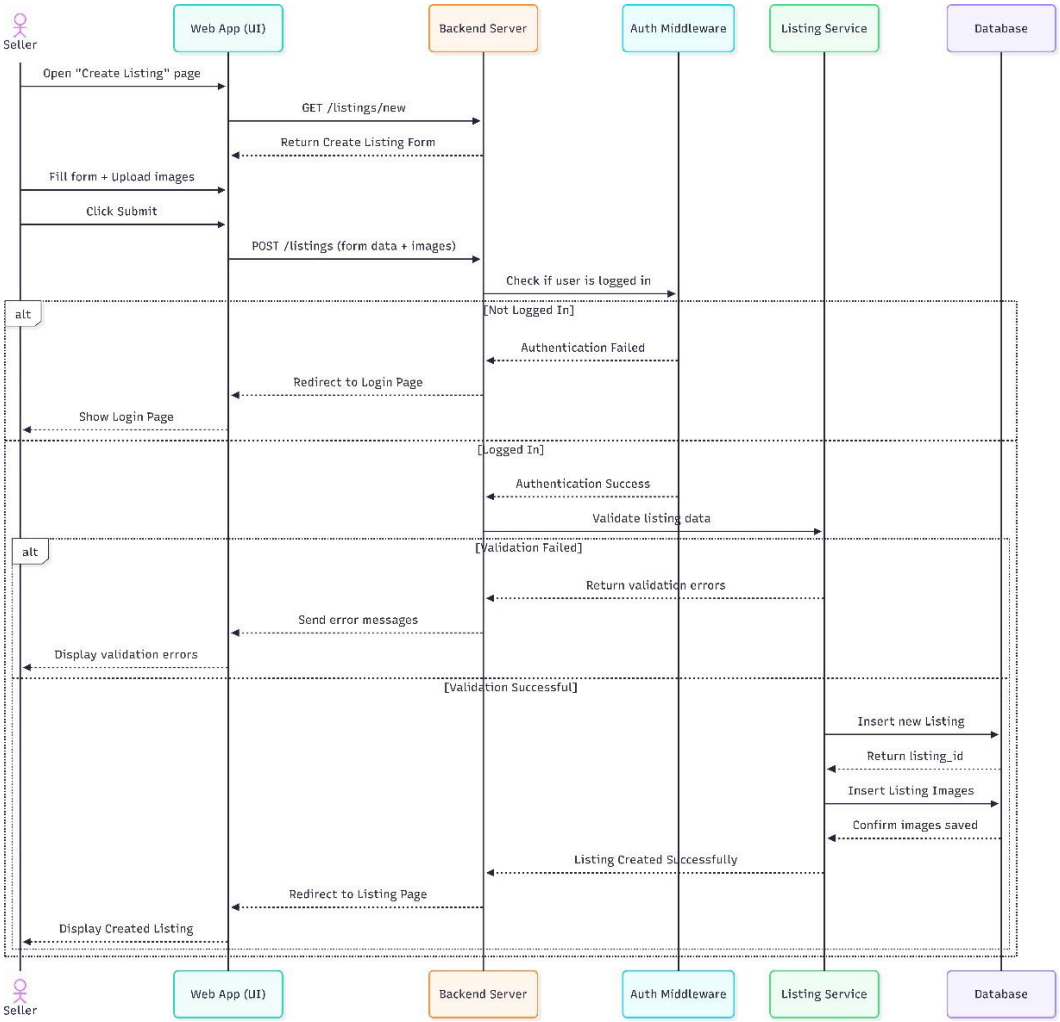
Decision nodes represent validation outcomes, ensuring both valid and invalid input paths are clearly defined. The diagram also models system error handling and redirect behaviour following successful completion.

This behavioural model ensures clarity in system logic before implementation begins.



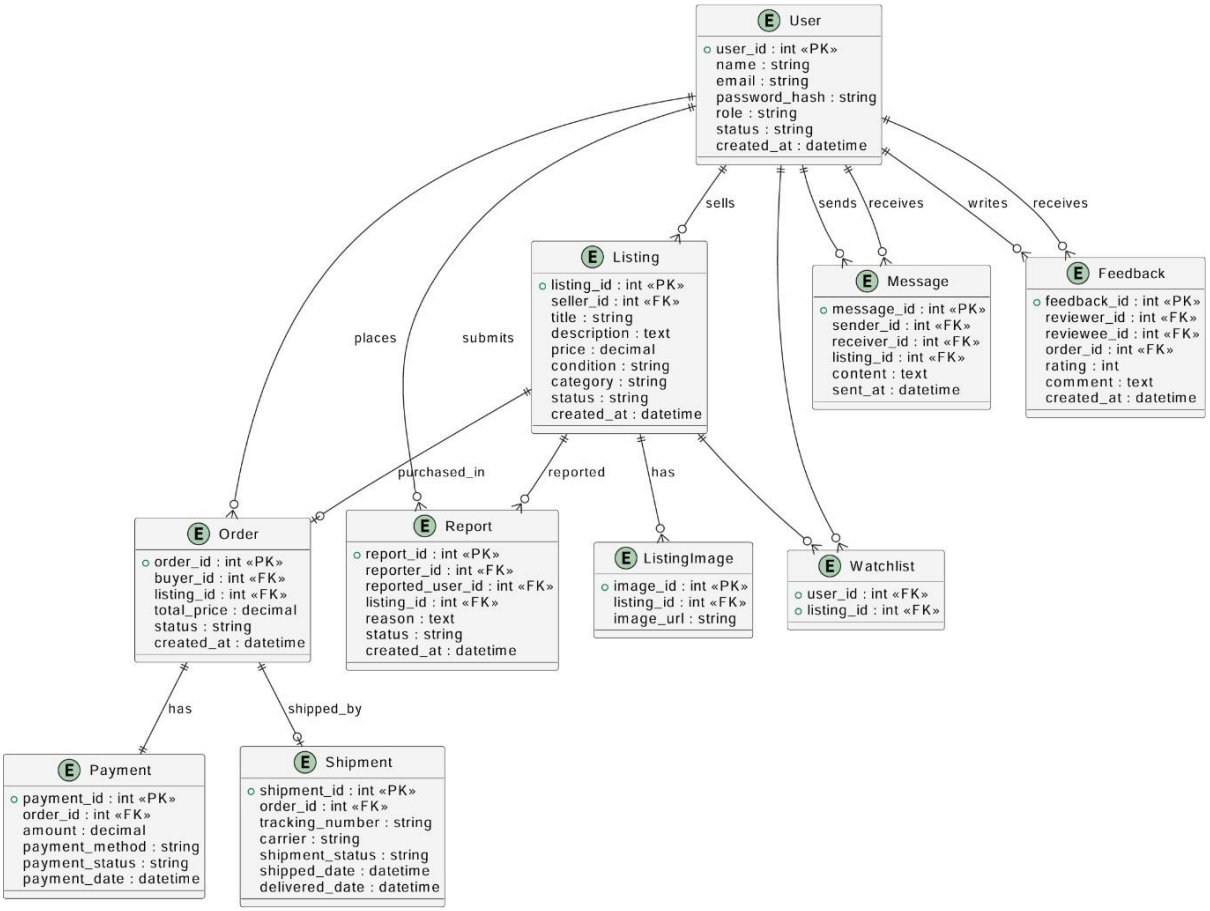
System Interaction Model – Sequence Diagram

The sequence diagram illustrates the interaction between system components during the listing creation process. It models communication between the web interface, backend server, authentication middleware, listing service layer, and database. Alternative interaction paths demonstrate authentication failure, validation errors, and successful data insertion. This diagram provides a dynamic representation of system behaviour and reinforces the separation of concerns within the architecture.



Database Design

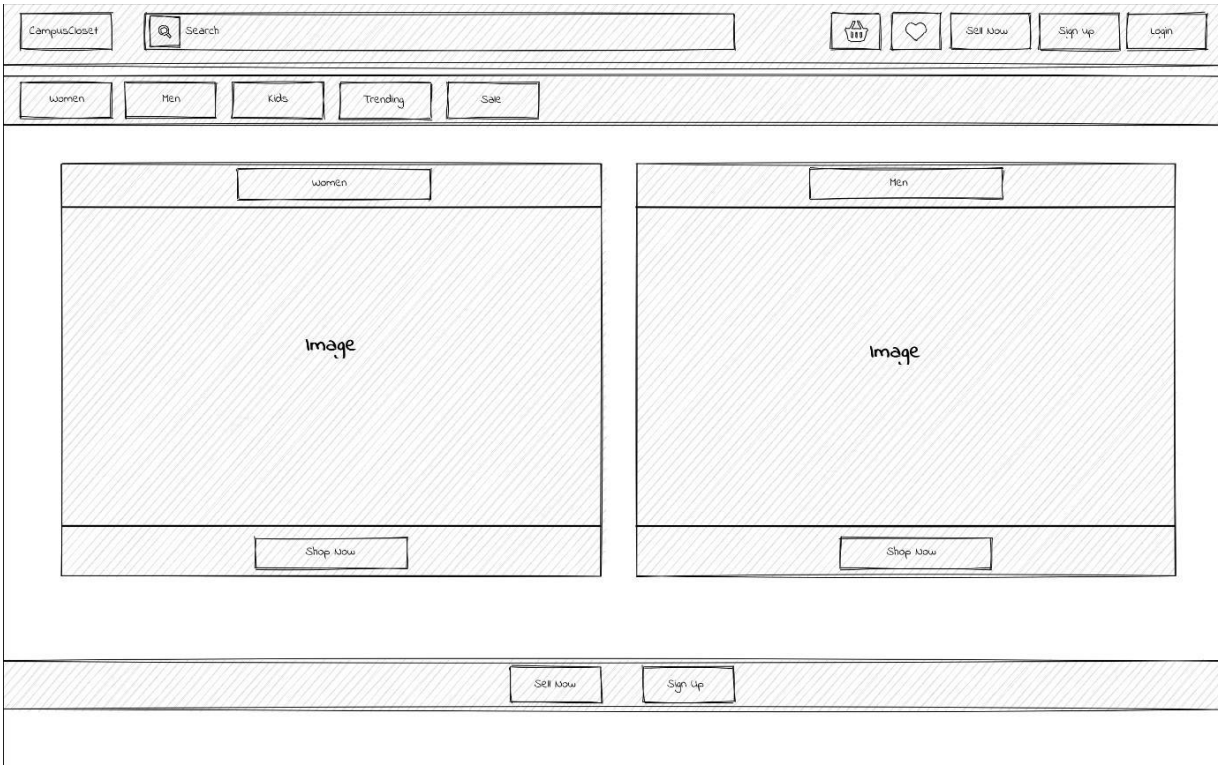
The database design defines the structural foundation of the ClosetSwap platform. The Entity Relationship Diagram models the entities required to support user management, listings, transactions, communication, and moderation.
Core entities include users, listings, listing images, orders, payments, shipments, messages, feedback, watchlists, and reports. Primary and foreign key relationships ensure referential integrity and support scalable system growth.
The relational structure allows for one-to-many and many-to-many relationships where appropriate, such as users owning multiple listings and users maintaining watchlists.
While the ERD represents the complete system scope, Sprint 3 will implement a subset of core tables to establish a stable, database-driven foundation before introducing advanced features.



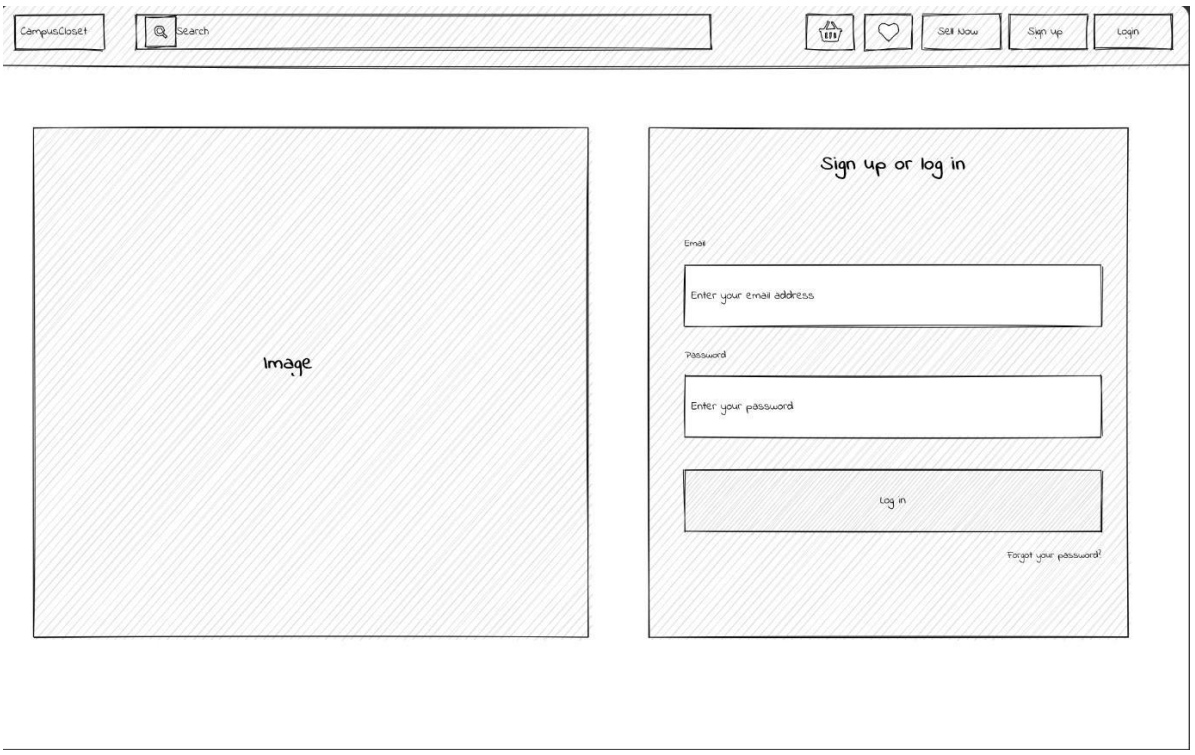
User Interface Planning – Wireframes

Wireframes were developed to visualise user interaction prior to implementation. These low-fidelity designs focus on layout, usability, and feature accessibility rather than visual styling.

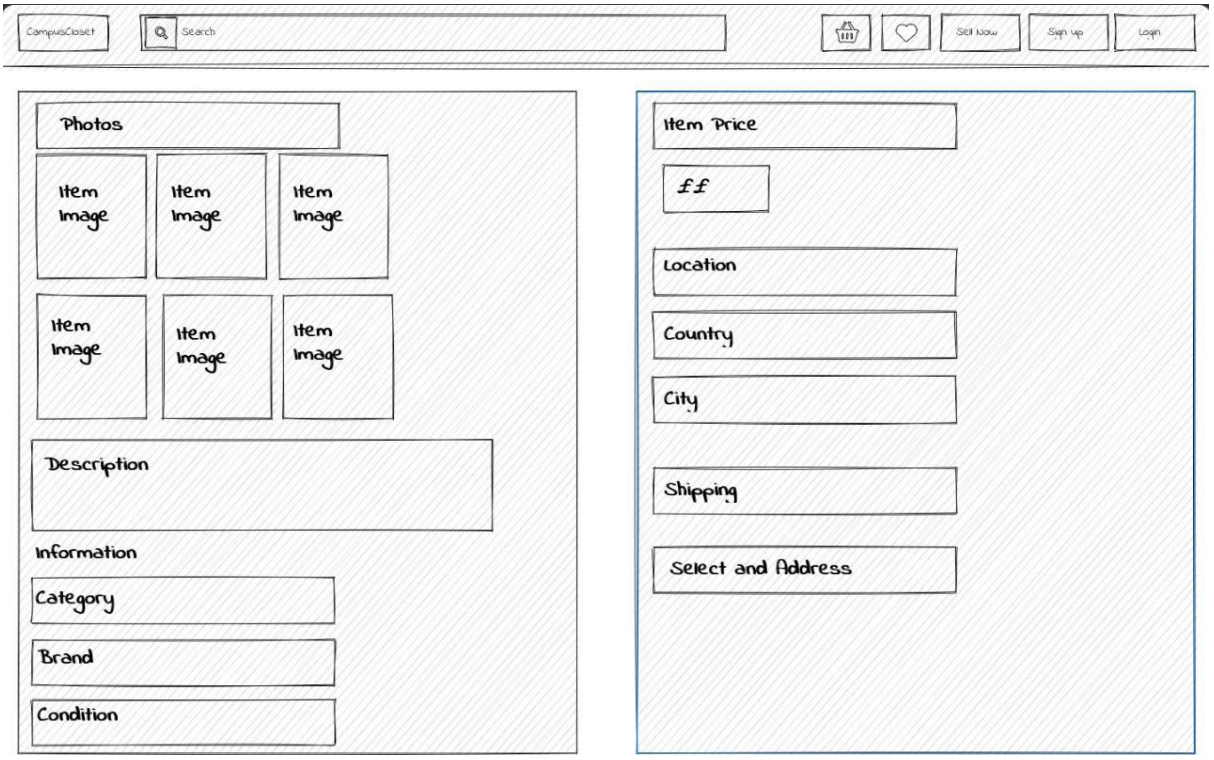
Home page:



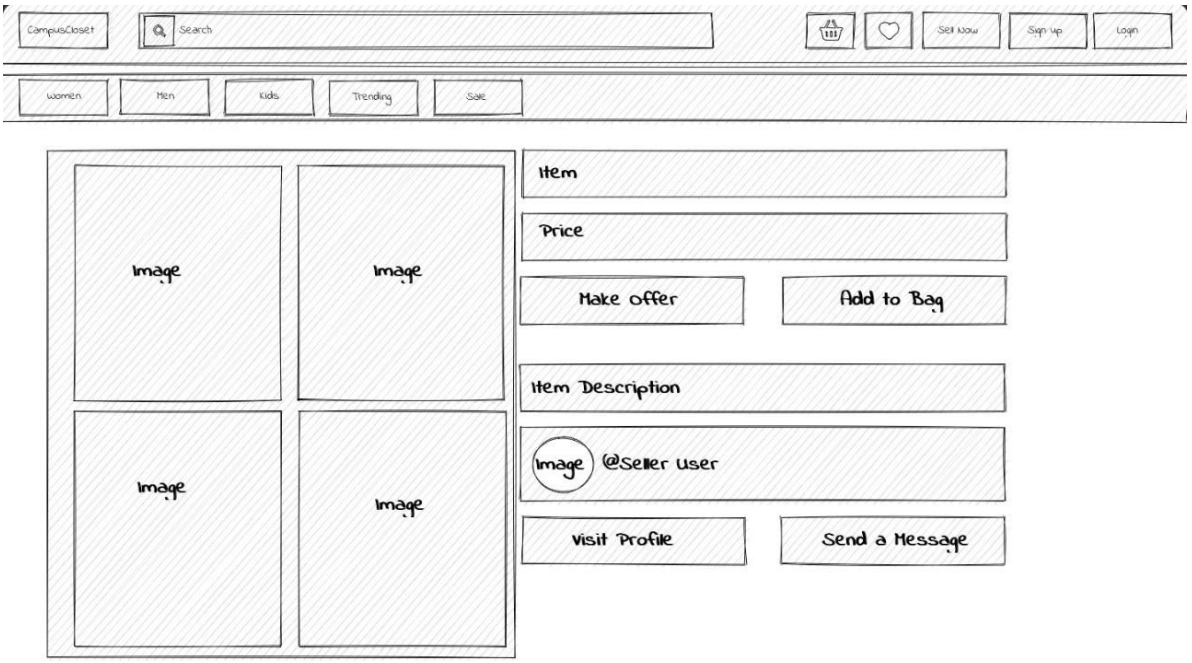
Sign up or Log in page:



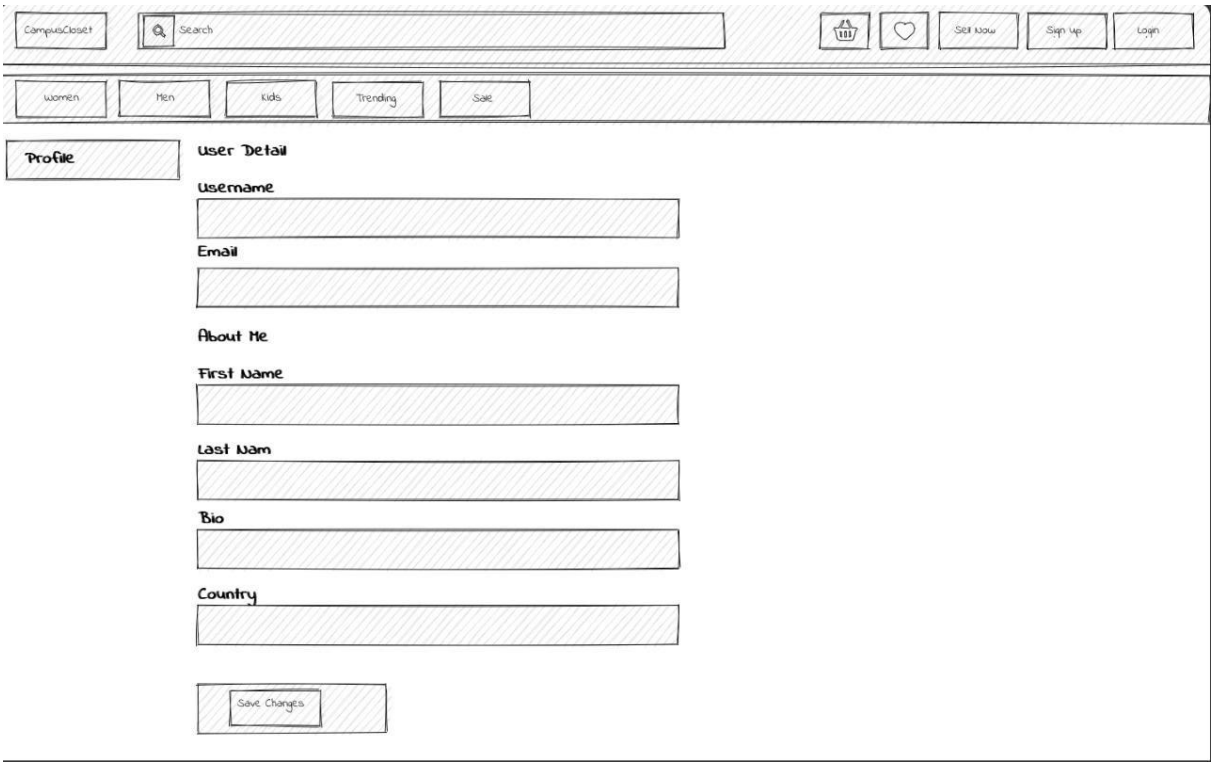
List item page:



View item page:



Profile page:



Sprint 3 Technical Specification

Sprint 3 marks the transition from modelling to implementation. The objective is to establish a functioning full-stack application using Node.js, Express.js, MySQL, PUG templating, and Docker.

The selected user stories for Sprint 3 focus on core functionality including user registration, login authentication, listing creation, listing editing and deletion, browsing functionality, and basic category filtering.

The scope of this sprint prioritises structural stability and CRUD functionality while intentionally excluding advanced features such as messaging, payment integration, watchlists, and recommendation systems. This controlled scope reduces technical risk and supports iterative development.

Planned routes include user authentication endpoints, listing CRUD endpoints, and category filtering endpoints. All routes will interact with the MySQL database and render dynamic content using PUG templates.

Task allocation within the team ensures clear responsibility across database design, authentication logic, listing functionality, templating, integration, and testing.

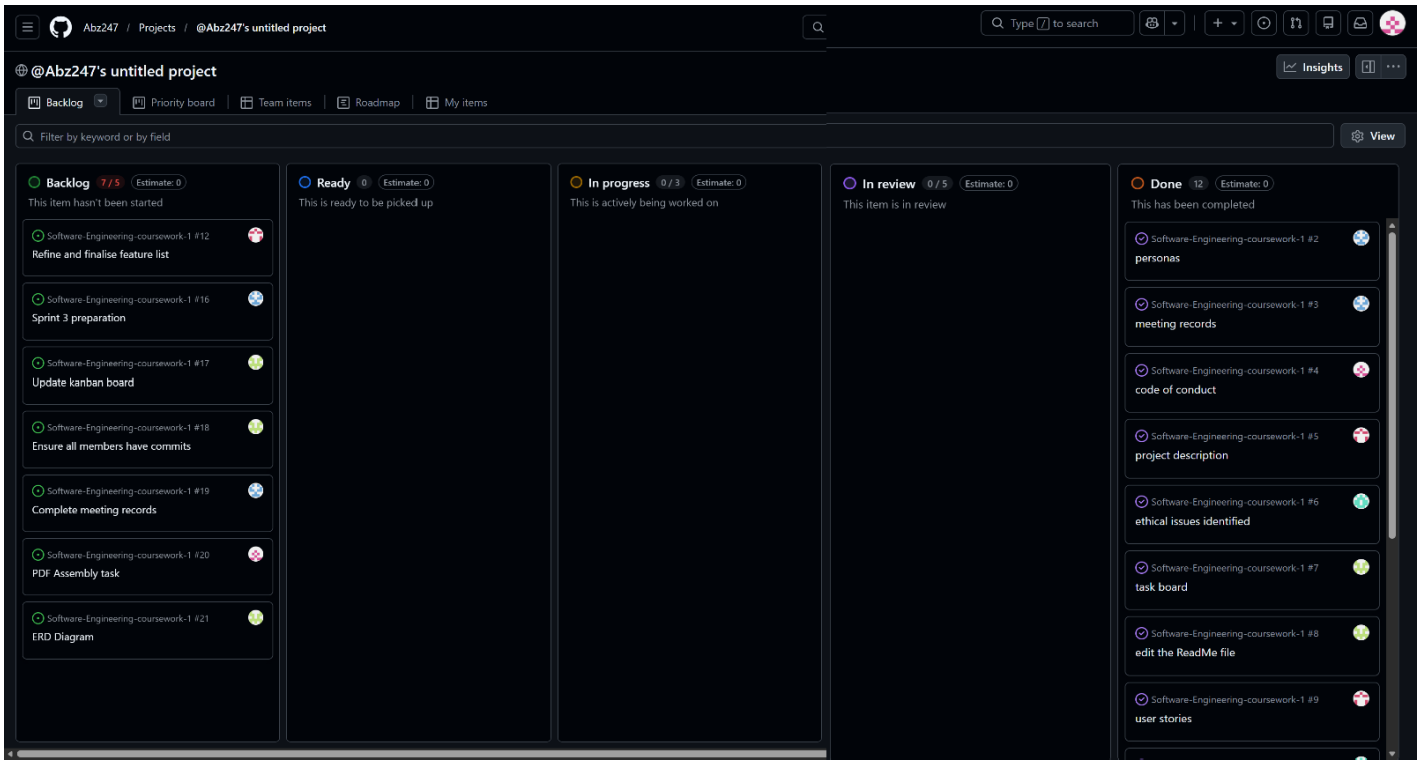
Risk mitigation strategies include frequent commits, structured branching, early integration testing, and regular sprint reviews.

See specification below:

Sprint 3 Technical Specification

Project Management and Evidence of Collaboration

The project is managed using GitHub Projects and a Kanban workflow. Tasks are categorised into backlog, in progress, review, and completed stages to ensure visibility of sprint progress.



The GitHub repository demonstrates collaborative contribution through commit history, pull requests, and issue tracking.

Repository Link:

[Abz247/Software-Engineering-coursework-1]

Contributions:

Regular meetings were conducted to review sprint objectives, allocate tasks, and resolve blockers. Meeting records provide documented evidence of decision-making and contribution tracking.
Meeting Records

Conclusion

Sprint 2 establishes a clear and structured foundation for development of the ClosetSwap platform. Through prioritised user stories, behavioural and structural modelling, database design, and scoped implementation planning, the system requirements have been translated into actionable technical specifications.
The project is prepared to transition into Sprint 3 development with defined architecture, controlled scope, and documented alignment between functional requirements and implementation strategy. This structured approach reflects adherence to Agile principles and supports iterative, risk-aware development.

User Stories

MUST HAVE (Core System Functionality)

These are essential for the platform to function.

User Story 1 – User Registration

Priority: MUST HAVE

User Story
As a prospective user, I want to register for an account so that I can access the platform's buying and selling features.

Acceptance Criteria

- **Given** a new visitor is on the registration page
- **When** they enter valid details (username, email, password)
- **Then** an account is created and stored securely in the database
- **Given** an email already exists
- **When** the user attempts to register
- **Then** the system displays an appropriate validation error

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User Story 2 – User Login & Authentication

Priority: MUST HAVE

User Story
As a registered user, I want to log in securely so that I can access my personal dashboard and manage listings.

Acceptance Criteria

- **Given** a registered user enters valid credentials
- **When** they submit the login form
- **Then** they are authenticated and redirected to their dashboard

- **Given** invalid credentials are entered
- **Then** an error message is displayed
- Passwords must be securely hashed in the database.

User Story 3 – Create Item Listing (CRUD – Create)

Priority: MUST HAVE

User Story
As a seller, I want to create a listing with images, description, size, condition, and price so that I can sell my used clothing items.

Acceptance Criteria

- **Given** a logged-in user
- **When** they complete the listing form and upload an image
- **Then** the item is stored in the database and visible to other users
- All required fields must be validated before submission.

User Story 4 – Edit / Delete Listing (CRUD – Update/Delete)

Priority: MUST HAVE

User Story
As a seller, I want to edit or remove my listings so that I can keep my items accurate and up to
date.

Acceptance Criteria

- **Given** the seller owns the listing
- **When** they edit the listing
- **Then** the updated information is saved
- **Given** the seller deletes the listing
- **Then** the item is removed from public view
- Users cannot modify listings they do not own.

User Story 5 – Browse Available Items (CRUD – Read)

Priority: MUST HAVE

User Story
As a user, I want to browse available items so that I can discover sustainable fashion products.

Acceptance Criteria

- **Given** a user visits the homepage
- **When** items exist in the database
- **Then** they are displayed in a responsive grid layout
- Each listing must show image, price, title, and seller.

SHOULD HAVE (Important but not critical)

User Story 6 – Search & Filter System

Priority: SHOULD HAVE

User Story
As a user, I want to search and filter items by category, size, brand, and price range so that I can quickly find relevant products.

Acceptance Criteria

- Users can enter keywords in a search bar
- Users can apply filters
- Results update dynamically or upon submission

User Story 7 – Secure Messaging System

Priority: SHOULD HAVE

User Story
As a buyer, I want to message sellers securely so that I can ask questions before making a purchase.

Acceptance Criteria

- Only logged-in users can send messages
- Messages are stored in the database
- Users can only view conversations they are part of

User Story 8 – Purchase Functionality

Priority: SHOULD HAVE

User Story
As a buyer, I want to purchase an item securely so that I can complete transactions safely.

Acceptance Criteria

- Users must be logged in to purchase
- Once purchased, the item is marked as sold
- The transaction is recorded in the database

COULD HAVE (Enhancement Features)

User Story 9 – Watchlist / Favourites

Priority: COULD HAVE

User Story
As a user, I want to save items to a favourites list so that I can easily return to them later.

Acceptance Criteria

- Users can add/remove items from favourites
- Favourited items are stored in the database
- Users can view all saved items in one page

User Story 10 – Transaction History

Priority: COULD HAVE

User Story
As a user, I want to view my purchase and sales history so that I can track my activity on the platform.

Acceptance Criteria

- Users can view a list of completed transactions
- Each record displays item name, date, and price

User Story 11 – User Profile Management

Priority: COULD HAVE

User Story
As a user, I want to update my profile details so that my account information remains accurate.

Acceptance Criteria

- Users can update username/email
- Password updates require confirmation
- Changes are reflected immediately

WON'T HAVE (For This Version)

User Story 12 – Dark Mode / Light Mode

Priority: WON'T HAVE (Phase 1)

User Story
As a user, I want to switch between dark and light themes so that I can customise my viewing experience.

Reason for Deprioritisation
While improving usability, this feature does not directly contribute to the core marketplace functionality and is therefore deferred to a later iteration.

Sprint 3 Specification Preparation

1. Sprint 3 Purpose

Sprint 3 marks the transition from planning and documentation into technical implementation.

The objective of this sprint is to develop the core database-driven foundation of the ClosetSwap web application using:

- Node.js
- Express.js
- MySQL
- PUG templating
- Docker containers

The focus of this sprint is to ensure that the application is dynamically connected to a database and capable of performing core CRUD operations. Rather than implementing all planned features, Sprint 3 prioritises structural stability, working routes, and proper database integration.

This approach ensures that future features in Sprint 4 can be built on a reliable and scalable foundation.

2. User Stories Selected for Sprint 3

The following user stories from the product backlog will be implemented during Sprint 3:

2.1 User Story 1 – User Registration (MUST HAVE)

As a prospective user, I want to register for an account so that I can access the platform.

Implementation:

- Registration form
- Server-side validation

- Password hashing before storage
- User stored securely in the users table
- Duplicate email prevention

2.2 User Story 2 – User Login (MUST HAVE)

As a registered user, I want to log in securely so that I can access my account.

Implementation:

- Login form
- Credential verification
- Basic session handling
- Error handling for invalid login attempts

2.3 User Story 3 – Create Item Listing (MUST HAVE)

As a seller, I want to create a listing with relevant details so that I can sell my used clothing.

Implementation:

- Create listing form
- Required field validation
- Data inserted into listings table
- Listing linked to logged-in user via foreign key

2.4 User Story 4 – Edit/Delete Listing (MUST HAVE)

As a seller, I want to edit or remove my listings so that they remain accurate.

Implementation:

- Edit listing functionality
- Delete listing functionality
- Ownership verification (users cannot modify listings they do not own)
- Updated information reflected immediately in database

2.5 User Story 5 – Browse Available Items (MUST HAVE)

As a user, I want to browse available items so that I can discover sustainable fashion products.

Implementation:

- Homepage displays listings dynamically
- Grid layout using PUG
- Each listing displays:
    - Title
    - Price
    - Image
    - Seller

2.6 User Story 6 – Basic Category Filtering (Partial)

Basic filtering by category will be implemented using route parameters.

Advanced filtering and keyword search will be implemented in Sprint 4.

3. Sprint 3 Scope Definition

Clear scope definition is critical to ensure realistic delivery and prevent scope creep.

3.1 Included in Sprint 3

- Fully implemented MySQL database schema
- Users list page
- User profile page
- Listings page
- Listing detail page
- Category filtering
- Listing CRUD operations
- Dynamic rendering using PUG
- Docker environment functioning for all members

- Evidence of GitHub contribution from all team members

3.2 Not Included in Sprint 3

The following features are intentionally excluded to maintain focus:

- Messaging system
- Purchase/payment functionality
- Watchlist/favourites
- Ratings and reviews
- Transaction history
- Advanced search system
- External API integrations

These features are planned for Sprint 4 once the core system is stable.

4. Database Design for Sprint 3

The database has been designed to support the implemented user stories while remaining extensible for future development.

4.1 Table: users

Field	Type	Description
id	INT (Primary Key)	Auto increment
username	VARCHAR	Unique username
email	VARCHAR	Unique email
password_hash	VARCHAR	Securely hashed password
created_at	DATETIME	Account creation

timestamp Purpose:

- Supports registration and login
- Links sellers to their listings
- Enables profile viewing

4.2 Table: categories

Field	Type	Description
id	INT (Primary Key)	Auto increment
name	VARCHAR	Category name (e.g. Jackets, Tops)

Purpose:

- Enables structured browsing
- Supports filtering functionality

4.3 Table: listings

Field	Type	Description
id	INT (Primary Key)	Auto increment
title	VARCHAR	Listing title
description	TEXT	Item
description		
price	DECIMAL	Item price
size	VARCHAR	Clothing size
condition	VARCHAR	Item
condition		
image_url	VARCHAR	Image reference
category_id	INT (Foreign Key)	Links to categories
user_id	INT (Foreign Key)	Links to users
status	VARCHAR	Available or sold
created_at	DATETIME	Timestamp

4.4 Relationships

- One user can create multiple listings.
- Each listing belongs to one user.
- Each listing belongs to one category.
- One category can contain multiple listings.

This relational design supports both browsing and filtering while maintaining data integrity.

5. Routes Implemented in Sprint 3

The following Express routes will be implemented:

5.1 User Routes

- GET /users
- GET /users/:id
- GET /register
- POST /register
- GET /login
- POST /login

5.2 Listing Routes

- GET /listings
- GET /listings/:id
- GET /listings/create
- POST /listings/create
- GET /listings/edit/:id
- POST /listings/edit/:id
- POST /listings/delete/:id

5.3 Category Route

- GET /categories/:id

All routes retrieve or manipulate real data from the MySQL database using the shared db.js service and render dynamic content using PUG templates.

6. Linking User Stories to Implementation

Each user story directly maps to both a database structure and a route:

User Story	Database Tables	Routes
Registration	users	/register
Login	users	/login
Create Listing	listings, users, categories	/listings/create
Edit/Delete	listings	/listings/edit/:id
Browse	listings, users	/listings
Filter	listings, categories	/categories/:id

This demonstrates traceability from requirement to implementation.

7. Task Allocation

To ensure accountability and equal contribution, tasks are allocated as follows:

Task	Assigned Member
Database Schema	Obaida
Authentication	Abdala
Listing CRUD	Meshari
PUG Templates	Eder
Docker & Integration	Abdullahi
Data Seeding & Testing	All Members

Contribution will be evidenced through GitHub commits and project board activity.

8. Risks and Mitigation

Potential risks during Sprint 3 include:

- Database connection errors
- Merge conflicts
- Authentication bugs
- Docker configuration issues

Mitigation strategies:

- Frequent commits and pull requests
- Clear branching strategy
- Early integration testing
- Regular stand-up communication

9. Sprint 3 Rationale

Sprint 3 has been deliberately scoped to prioritise core functionality over advanced features. By focusing on database integration, CRUD operations, and dynamic rendering, the team ensures that the system architecture is stable before introducing additional complexity.

This reflects an iterative Agile approach, where value is delivered incrementally and validated at each sprint review.

Establishing a working full-stack foundation in Sprint 3 reduces technical risk and allows Sprint 4 to focus on feature expansion rather than structural corrections.

Meeting Minutes

Date and Time	Date: 9th February 2026 Time: 13:30
Project Name	ClosetSwap
Meeting Goal	• Review and refine user stories • Confirm feature prioritisation (MoSCoW) • Plan diagram responsibilities • Begin structuring Sprint 3 preparation
Facilitator	Obaida Muqbel
Note taker	Meshari Alsadoun
Attendees	Meshari Alsadoun Obaida Muqbel Abdala Mohamed Abdullahi Mahamud Eder Siva
Roundtable Updates (each group member to contribute)	Eder presented the drafted user stories and explained the logic behind feature prioritisation. Abdala discussed initial wireframe concepts and how they align with the must have stories.

	Abdullahi outlined early thoughts on database structure and required entities. Meshari explained the purpose of Sprint 3 specification preparation and proposed structuring it around selected user stories. Obaida raised points about ensuring diagrams clearly reflect system flow and validation logic.
Discussion points	The group reviewed and refined all user stories to ensure clarity and measurable acceptance criteria. MoSCoW prioritisation was confirmed, with messaging and payment features categorised outside Sprint 3 scope. Responsibilities were agreed: Eder – User Stories Abdala – Wireframes and Use Case Diagram Abdullahi – ERD and GitHub leader Meshari – Sprint 3 Specification Obaida – Sequence Diagram and Activity Diagram and PDF compilation The importance of aligning diagrams with actual implementation plans was highlighted.
Actions (list tasks and assign a group member)	Eder to finalise and format the user stories document. Abdala to complete wireframes for homepage, login, listing creation and item detail. Obaida to begin drafting activity and sequence diagrams. Abdullahi to draft ERD structure. Meshari to outline Sprint 3 specification framework. All members to update GitHub project board.

Meeting Minutes

Date and Time	Date: 16th February 2026 Time: 14:30
Project Name	ClosetSwap
Meeting Goal	• Review completed diagrams • Finalise ERD design • Align Sprint 3 specification with modelling artefacts • Ensure scope consistency
Facilitator	Obaida Muqbel
Note taker	Meshari Alsadoun
Attendees	Meshari Alsadoun Obaida Muqbel Abdala Mohamed Abdullahi Mahamud Eder Siva
Roundtable Updates (each group member to contribute)	Eder confirmed the final version of user stories was complete and aligned with the prioritisation agreed previously. Abdala presented the completed wireframes and use case diagram, explaining actor relationships and system boundary. Obaida presented the activity diagram for listing creation and

	sequence diagram for system interaction. Abdullahi presented the ERD draft and explained entity relationships and foreign keys. Meshari shared progress on the Sprint 3 specification document and how selected user stories map to routes and database tables.
Discussion points	The team reviewed consistency between user stories and diagrams to ensure alignment. The ERD was adjusted to clarify relationships between users, listings, and orders. It was agreed that Sprint 3 would focus only on core tables (users, listings, categories) despite the ERD modelling full system scope. The group confirmed that messaging, checkout, and advanced features would remain excluded from Sprint 3 implementation.
Actions (list tasks and assign a group member)	Abdullahi to finalise the ERD diagram. Obaida to refine activity and sequence diagrams for clarity. Abdala to ensure wireframes align with selected Sprint 3 features only. Meshari to complete the Sprint 3 specification document. All members to review artefacts before final meeting.

Meeting Minutes

Date and Time	Date: 23rd February 2026 Time: 13:30
Project Name	ClosetSwap
Meeting Goal	• Final review of Sprint 2 deliverables • Confirm all required checklist items are included • Prepare final PDF structure • Review GitHub contributions
Facilitator	Obaida Muqbel
Note taker	Meshari Alsadoun
Attendees	Meshari Alsadoun Obaida Muqbel Abdala Mohamed Abdullahi Mahamud Eder Siva
Roundtable Updates (each group member to contribute)	Eder confirmed user stories were complete and correctly formatted for submission. Abdala reviewed wireframes and use case diagram placement within the final document.

	Obaida confirmed sequence and activity diagrams were finalised. Abdullahi verified ERD clarity and confirmed database relationships were correct. Meshari confirmed Sprint 3 specification preparation was complete and aligned with selected features.
Discussion points	The group cross-checked the Sprint 2 checklist to ensure inclusion of: User stories, use case diagram, activity diagram, wireframes, Sprint 3 specification, ERD, Kanban screenshot, GitHub link, and meeting records. GitHub contributions were reviewed to ensure all members had committed work. The document structure was agreed to follow a logical progression from requirements to modelling to specification. Final formatting consistency and diagram clarity were discussed.
Actions (list tasks and assign a group member)	Obaida to merge all documents into a single PDF. Abdullahi to confirm GitHub link functionality. Obaida to upload final submission. All members to review final version before submission deadline.